

INDR 100

INTRODUCTION TO INDUSTRIAL ENGINEERING

Lecture #14 Work Sampling

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1

Agenda of the Day

- ❑ Announcement
- ❑ Work Sampling
- ❑ Worker Compensation

INDR 100

2

2

Announcement

- ❑ Midterm will be held on 25 April 2026, Saturday, between 11:45-14:45.
- ❑ Venue will be ENG Z50.
- ❑ You will be seated according to a plan.
- ❑ Only essential materials (pencil, eraser, calculator, ID and water) will be allowed. All belongings should be left at the podium of the classroom.
- ❑ No phones, watches or any digital equipment are allowed.

INDR 100

3

3

Announcement – cont'd

- ❑ It will be a closed-book and closed-notes exam.
- ❑ No cheat-sheet is necessary.
- ❑ You are not allowed to ask any questions during the exam.
- ❑ Duration of the exam will be around 120 minutes. It will not be permitted to leave and come back during the exam.

INDR 100

4

4

Announcement – cont'd

- ❑ Midterm will include all topics we covered till the exam date.
- ❑ There will be different types of questions:
 - Quantitative: methodology of IE.
 - Qualitative: concepts of IE.
 - Questions will be in the format of test questions: fill-in-the-blanks, true-false
- ❑ You must show your work clearly and write legibly to receive full credit.
- ❑ There are a set of study questions uploaded at LearnHub.

INDR 100

5

5

Work Sampling

- ❑ Technique for estimating the proportion of time that a worker or machine spends on various activities
- ❑ Observers make *brief observations* of a worker or a machine at *random intervals* over a *period of time* and simply note the nature of the activity
- ❑ Purpose:
 - To estimate percentage of unproductive or idle time for repetitive jobs
 - To estimate the percentage of time spent on various tasks for non-repetitive jobs

INDR 100

6

6

Work Sampling

- ❑ Work sampling is better suited to jobs involving non-repetitive long cycles.
- ❑ Job is expected to be changed periodically as customer orders change.
- ❑ Job produces relatively small quantities of output.
- ❑ Resulting time standard used for accounting cost standard, pricing analysis, and production planning.

INDR 100

7

7

Work Sampling steps

- ❑ Determine the sample size
- ❑ Train the observers
- ❑ Develop random sample schedule
- ❑ Take observations, and re-compute the desired sample size several times if initial estimates are not reliable
- ❑ Determine the estimated proportion of time spent on specified activity

INDR 100

8

8

The statistical formula for sample size

$$N = \frac{Z^2 \cdot p \cdot (1 - p)}{E^2}$$

Where:

- N : Total number of random observations required.
- Z : Standard normal statistic for the confidence level (see table below).
- p : Estimated proportion of the activity (from preliminary study or history).
- E : Absolute error (e.g., $\pm 4\%$ means $E = 0.04$).

Confidence Level	Z-Value
90%	1.645
95%	1.96
99%	2.58

9

Scenario: A quality engineer performs a preliminary study ($N = 50$) and estimates that the proportion of time spent on "Productive Work" is **80%** ($p = 0.80$).

Goal: Calculate the minimum number of observations (N) required to be **95% confident** that the observed proportion is within $\pm 4\%$ ($E = 0.04$).

Solution:

1. Identify the parameters:

- $p = 0.80$
- $Z = 1.96$ (for 95% confidence)
- $E = 0.04$

2. Substitute into the formula:

$$N = \frac{(1.96)^2 \cdot 0.80 \cdot (1 - 0.80)}{(0.04)^2}$$

3. Calculate:

$$N = \frac{3.8416 \cdot 0.80 \cdot 0.20}{0.0016}$$

$$N = \frac{3.8416 \cdot 0.16}{0.0016}$$

$$N = \frac{0.614656}{0.0016} \approx 384.16$$

4. **Conclusion:** Always round up to the next whole integer.

$$N = 385 \text{ observations}$$

10

Example-1: Work Sampling

A work sampling study was performed on an electronic assembly operation at OK Instruments. The study covered an 8-hour shift with a single worker.

Minutes after start	Assemble units	Allowances
4	1	0
13	1	0
20	1	0
21	1	0
25	1	0
30	1	0
48	1	0
63	1	0
80	1	0
94	1	0
120	1	0
126	1	0
127	1	0
128	1	0
146	1	0
160	1	0
170	1	0
173	0	1
177	1	0
184	0	1

INDR 100

Minutes after start	Assemble units	Allowances
200	0	1
224	1	0
229	0	1
230	1	0
250	1	0
252	1	0
256	1	0
286	1	0
303	1	0
309	1	0
337	1	0
343	1	0
351	1	0
360	1	0
377	1	0
378	1	0
412	1	0
430	0	1
452	1	0
453	1	0
Total	35	5
Average	0.875	0.125

11

11

Example-1: Work Sampling

A work sampling study was performed on an electronic assembly operation at OK Instruments. The study covered an 8-hour shift with a single worker. The results of the study were:

<u>Activity</u>	<u>% of Worker's Time</u>
Assemble Units	87.5
Allowances	12.5

If the worker received a performance rating of 1.20 on the Assemble Units activity and 400 units were assembled during the study, what is the labor standard for this operation?

INDR 100

12

12

Example-1: Work Sampling

- 1) Compute the average time per assemble:

$$\frac{\text{Total Minutes of Assembly Work}}{\text{Number of Units Assembled}} \\ = 0.875(480)/400 = 1.05 \text{ minutes per unit}$$

- 2) Compute the normal time per unit:

$$= (\text{Average Time per Unit}) (\text{Performance Rating}) \\ = 1.05 (1.20) = 1.26 \text{ minutes per unit}$$

- 3) Compute the labor standard:

$$= \text{Normal Time} * (1 + \text{Allowance}) \\ = 1.26 (1 + 0.125) = 1.4175 \text{ minutes per unit}$$

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13

13

Example-2 Work Sampling

- You want to check whether the worker spends too much time on **non-value-added tasks (walking, searching)**.
- You perform **200 random observations** throughout the day and find:
 - Worker packing: **120 observations**
 - Worker walking/searching: **60 observations**
 - Worker idle/waiting: **20 observations**
- **Questions**
 - What percentage of the time is the worker doing value-added work?
 - Do the walking + idle times justify a method improvement study?

14

Example-2 Work Sampling

Solution

- Value-added (packing):
 - $120/200 = 60\%$
- Walking/searching:
 - $60/200 = 30\%$
- Idle:
 - $20/200 = 10\%$
- Only **60% of the time** is spent adding value.
The **40% non-value-added time** strongly suggests improvement opportunities.

15

How to carry out:

- Choose random times to observe an employee.
 - Start at 8:30 am, which corresponds to 0.
 - Finish at 17:30, which corresponds to 1.
 - Skip lunch time at 12:00-13:00.
 - Create a random variable in (0,1), say 0.345. Multiply it by $8 \times 60 = 480$, so approximately 166 minutes (corresponds to 2 hours and 46 minutes).
 - Check what the employee is doing at 11:16, and record it.
- Record what the employee is doing:
 - e.g., a bank employee: working on a document (activity A), assisting a customer (activity B), attending a personal need (activity C) etc.
- Percent occurrence for each activity can be calculated.
 - e.g. percent for activity A=?
 - What can go wrong?
 - Which issues should we be careful about?

INDR 100

16

16

Disadvantages of Work Sampling

- ❑ Less detail on elements/tasks of a job
- ❑ Workers may alter patterns
- ❑ Often no record of method used by worker
- ❑ Observers may fail to adhere to random observation schedule
- ❑ Not useful for short, repetitive tasks
- ❑ Much time required to move from observation area to observation area to ensure randomness

INDR 100

17

17

Advantages of Work Sampling

- ❑ Observations less susceptible to short term fluctuations
- ❑ Little or no work disruption
- ❑ Workers are less resentful
- ❑ Less costly and time-consuming
- ❑ Many studies can be conducted simultaneously
- ❑ Useful for non-repetitive tasks

INDR 100

18

18

Worker Compensation

- ❑ Compensation schemes:
 - Time based: pay based on the number of hours worked -- most common in healthcare
 - Output based: pay based on the number of units completed -- more difficult to operationalize, yet pay is related to efforts
- ❑ Incentive Systems
 - Profit sharing plans -- receive % of profits
 - Gain sharing plans -- receive a % of the value (i.e., cost savings) realized through increases in productivity

INDR 100

19